

PRACTICAL 3

```
mysql> USE my_database;
Database changed
mysql> CREATE TABLE orders (
  ->     ord_no INT PRIMARY KEY,
  ->     purch_amt DECIMAL(10,2),
  ->     ord_date DATE,
  ->     customer_id INT,
  ->     salesman_id INT
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_my_database |
+-----+
| orders                 |
| student                |
+-----+
2 rows in set (0.00 sec)
```

```
mysql> INSERT INTO orders VALUES
  -> (70001,150.50,'2012-10-05',3005,5002),
  -> (70009,270.65,'2012-09-10',3001,5005),
  -> (70002,65.26,'2012-10-05',3002,5001),
  -> (70004,110.50,'2012-08-17',3009,5003),
  -> (70007,948.50,'2012-09-10',3005,5002),
  -> (70005,2400.60,'2012-07-27',3007,5001),
  -> (70008,5760.00,'2012-09-10',3002,5001),
  -> (70010,1983.43,'2012-10-10',3004,5006),
  -> (70003,2480.40,'2012-10-10',3009,5003),
  -> (70012,250.45,'2012-06-27',3008,5002),
  -> (70011,75.29,'2012-08-17',3003,5007),
  -> (70013,3045.60,'2012-04-25',3002,5001);
Query OK, 12 rows affected (0.01 sec)
Records: 12  Duplicates: 0  Warnings: 0

mysql>
mysql> SELECT * FROM orders;
+-----+-----+-----+-----+-----+
| ord_no | purch_amt | ord_date | customer_id | salesman_id |
+-----+-----+-----+-----+-----+
| 70001 | 150.50 | 2012-10-05 | 3005 | 5002 |
| 70002 | 65.26 | 2012-10-05 | 3002 | 5001 |
| 70003 | 2480.40 | 2012-10-10 | 3009 | 5003 |
| 70004 | 110.50 | 2012-08-17 | 3009 | 5003 |
| 70005 | 2400.60 | 2012-07-27 | 3007 | 5001 |
| 70007 | 948.50 | 2012-09-10 | 3005 | 5002 |
| 70008 | 5760.00 | 2012-09-10 | 3002 | 5001 |
| 70009 | 270.65 | 2012-09-10 | 3001 | 5005 |
| 70010 | 1983.43 | 2012-10-10 | 3004 | 5006 |
| 70011 | 75.29 | 2012-08-17 | 3003 | 5007 |
| 70012 | 250.45 | 2012-06-27 | 3008 | 5002 |
| 70013 | 3045.60 | 2012-04-25 | 3002 | 5001 |
+-----+-----+-----+-----+-----+
12 rows in set (0.00 sec)
```

```
mysql> SELECT *
  -> FROM orders
  -> where purch_amt > 2000;
```

ord_no	purch_amt	ord_date	customer_id	salesman_id
70003	2480.40	2012-10-10	3009	5003
70005	2400.60	2012-07-27	3007	5001
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT ord_no, purch_amt
  -> FROM orders
  -> WHERE purch_amt > 2000;
```

ord_no	purch_amt
70003	2480.40
70005	2400.60
70008	5760.00
70013	3045.60

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT *
  -> FROM orders
  -> WHERE ord_date = '2012-09-10';
```

ord_no	purch_amt	ord_date	customer_id	salesman_id
70007	948.50	2012-09-10	3005	5002
70008	5760.00	2012-09-10	3002	5001
70009	270.65	2012-09-10	3001	5005

```
3 rows in set (0.03 sec)
```

```
mysql> SELECT ord_no, purch_amt
  -> FROM orders
  -> WHERE ord_date = '2012-09-10';
```

ord_no	purch_amt
70007	948.50
70008	5760.00
70009	270.65

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM orders
-> WHERE salesman_id = 5001;
```

ord_no	purch_amt	ord_date	customer_id	salesman_id
70002	65.26	2012-10-05	3002	5001
70005	2400.60	2012-07-27	3007	5001
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001

4 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM orders
-> ORDER BY purch_amt DESC;
```

ord_no	purch_amt	ord_date	customer_id	salesman_id
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
70003	2480.40	2012-10-10	3009	5003
70005	2400.60	2012-07-27	3007	5001
70010	1983.43	2012-10-10	3004	5006
70007	948.50	2012-09-10	3005	5002
70009	270.65	2012-09-10	3001	5005
70012	250.45	2012-06-27	3008	5002
70001	150.50	2012-10-05	3005	5002
70004	110.50	2012-08-17	3009	5003
70011	75.29	2012-08-17	3003	5007
70002	65.26	2012-10-05	3002	5001

12 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM orders
-> ORDER BY ord_date;
```

ord_no	purch_amt	ord_date	customer_id	salesman_id
70013	3045.60	2012-04-25	3002	5001
70012	250.45	2012-06-27	3008	5002
70005	2400.60	2012-07-27	3007	5001
70004	110.50	2012-08-17	3009	5003
70011	75.29	2012-08-17	3003	5007
70007	948.50	2012-09-10	3005	5002
70008	5760.00	2012-09-10	3002	5001
70009	270.65	2012-09-10	3001	5005
70001	150.50	2012-10-05	3005	5002
70002	65.26	2012-10-05	3002	5001
70003	2480.40	2012-10-10	3009	5003
70010	1983.43	2012-10-10	3004	5006

12 rows in set (0.00 sec)

```
mysql> SELECT SUM(purch_amt) AS toal_revenue  
-> FROM orders;
```

```
+-----+  
| toal_revenue |  
+-----+  
|      17541.18 |  
+-----+  
1 row in set (0.00 sec)
```

```
mysql> SELECT AVG(purch_amt) AS average_order  
-> FROM orders;
```

```
+-----+  
| average_order |  
+-----+  
|    1461.765000 |  
+-----+  
1 row in set (0.00 sec)
```

```
mysql> SELECT MAX(purch_amt) AS highest_order  
-> FROM orders;
```

```
+-----+  
| highest_order |  
+-----+  
|        5760.00 |  
+-----+  
1 row in set (0.00 sec)
```

```
mysql> SELECT MIN(purch_amt) AS lowest_order  
-> FROM orders;
```

```
+-----+  
| lowest_order |  
+-----+  
|         65.26 |  
+-----+  
1 row in set (0.00 sec)
```

```
mysql> SELECT COUNT(*) AS total_orders  
-> FROM orders;
```

```
+-----+  
| total_orders |  
+-----+  
|           12 |  
+-----+  
1 row in set (0.01 sec)
```

```
mysql> SELECT salesman_id,  
-> SUM(purch_amt) AS total_sales  
-> FROM orders  
-> GROUP BY salesman_id;
```

salesman_id	total_sales
5002	1349.45
5001	11271.46
5003	2590.90
5005	270.65
5006	1983.43
5007	75.29

6 rows in set (0.00 sec)

```
mysql>  
mysql>  
mysql> SELECT customer_id,  
-> SUM(purch_amt) AS total_purchase  
-> FROM orders  
-> GROUP BY customer_id;
```

customer_id	total_purchase
3005	1099.00
3002	8870.86
3009	2590.90
3007	2400.60
3001	270.65
3004	1983.43
3003	75.29
3008	250.45

8 rows in set (0.00 sec)

```
mysql> SELECT customer_id,  
-> MAX(purch_amt) AS highest_purchase  
-> FROM orders  
-> GROUP BY customer_id;
```

customer_id	highest_purchase
3005	948.50
3002	5760.00
3009	2480.40
3007	2400.60
3001	270.65
3004	1983.43
3003	75.29
3008	250.45

8 rows in set (0.00 sec)

```
mysql> SELECT salesman_id,  
-> SUM(purch_amt) AS total_sales  
-> FROM orders  
-> GROUP BY salesman_id  
-> HAVING SUM(purch_amt) > 3000;
```

salesman_id	total_sales
5001	11271.46

1 row in set (0.00 sec)

```
mysql> SELECT customer_id,
->          SUM(purch_amt) AS total_purchase
-> FROM orders
-> GROUP BY customer_id
-> HAVING SUM(purch_amt) > 2500;
```

customer_id	total_purchase
3002	8870.86
3009	2590.90

2 rows in set (0.00 sec)

```
mysql> SELECT customer_id,
->          COUNT(*) AS total_orders
-> FROM orders
-> GROUP BY customer_id
-> HAVING COUNT(*) > 1;
```

customer_id	total_orders
3005	2
3002	3
3009	2

3 rows in set (0.00 sec)

```
mysql> SELECT customer_id,
->          SUM(purch_amt) AS total_purchase
-> FROM orders
-> GROUP BY customer_id
-> HAVING SUM(purch_amt) > 1000
-> ORDER BY total_purchase DESC;
```

customer_id	total_purchase
3002	8870.86
3009	2590.90
3007	2400.60
3004	1983.43
3005	1099.00

5 rows in set (0.00 sec)

BETWEEN Operator

BETWEEN is used to select values that fall within a specified range, including the boundary values.

```
mysql> SELECT customer_id,  
-> MAX(purch_amt) AS max_purchase  
-> FROM orders  
-> GROUP BY customer_id  
-> HAVING MAX(purch_amt) BETWEEN 2000 AND 6000;
```

customer_id	max_purchase
3002	5760.00
3009	2480.40
3007	2400.60

3 rows in set (0.00 sec)

```
mysql> SELECT salesman_id,  
-> COUNT(*) AS total_orders  
-> FROM orders  
-> GROUP BY salesman_id  
-> HAVING COUNT(*) >= 2;
```

salesman_id	total_orders
5002	3
5001	4
5003	2

3 rows in set (0.00 sec)

```
mysql> SELECT ord_date,  
-> MAX(purch_amt) AS highest_purchase  
-> FROM orders  
-> GROUP BY ord_date  
-> HAVING MAX(purch_amt) > 2000;
```

ord_date	highest_purchase
2012-10-10	2480.40
2012-07-27	2400.60
2012-09-10	5760.00
2012-04-25	3045.60

4 rows in set (0.00 sec)

- WHERE → Filters rows before grouping.
- GROUP BY → Creates groups of rows.
- HAVING → Filters groups after grouping.
- ORDER BY → Sorts the final result.

